

Paper v1 manuscript

Repository source: `paper/README.md`

Status

This directory contains the lean Paper v1 manuscript for the proposed exact universal contact-geometric lower bound

$$\text{Vol}(K) \geq \frac{26220940089713\pi}{200000000000000} d^3 = 0.41187756378030227424817892598 \dots d^3$$

for every three-dimensional convex body K of constant width d .

The manuscript is an **unreleased computer-assisted research draft**. Its current author line is provisional pending final authorship and affiliation sign-off. It is not a peer-reviewed publication, does not establish sharpness or equality, does not identify a minimizer, and does not prove the Meissner conjecture.

Files

- `contact_geometric_bound.tex` is the editable LaTeX manuscript.
- `contact_geometric_bound.pdf` is the checksum-protected reference PDF.
- `reconcile_certificate.py` reads the authoritative certificate JSON and generates/checks the theorem-critical TeX includes.
- `certificate_constants.tex` is the generated exact-constant include.
- `lower_branch_table.tex` is the generated gap-free lower-branch table.
- `build.sh` performs a non-destructive local build in `paper/build/`.
- `SHA256SUMS.txt` authenticates the archived paper sources and reference PDF.

The reference PDF is committed for review. A local build is written to `paper/build/contact_geometric_bound.pdf` and does not overwrite the reference artifact.

Certificate reconciliation

The paper does not maintain independent hand-entered copies of the main coefficient, circumradius split, lower-branch table, fan census, core and cap bounds, near-Jung lower bound, or handoff margin. Those values are generated from

```
certificate/certificates/universal_contact_bound_certificate.json
```

Run the reconciliation check from the repository root:

```
python3 paper/reconcile_certificate.py --check
```

The required marker is:

```
MINVOL PAPER-CERTIFICATE RECONCILIATION: PASS
```

To deliberately regenerate the two TeX includes after an authorized certificate change:

```
python3 paper/reconcile_certificate.py --write
```

A certificate change is a mathematical change and requires explicit review, new checksums, and a new public version. Ordinary paper editing should use `--check`, not `--write`.

Build

Required tools are Python 3, `latexmk`, and a PDFLaTeX installation containing the standard AMS, Latin Modern, `microtype`, `geometry`, `hyperref`, `xurl`, `enumitem`, and `table` packages used in the source.

From the repository root:

```
./build_paper.sh
```

Expected final markers:

```
MINVOL PAPER-CERTIFICATE RECONCILIATION: PASS  
MINVOL PAPER BUILD: PASS
```

The build uses a fixed source-date epoch and canonicalizes the PDF trailer ID. Equivalent PDF bytes are nevertheless not required across all TeX engines, font maps, operating systems, or compression libraries. The mathematical proof authority is the exact certificate and verifier, not PDF serialization.

Manuscript boundary

Paper v1 explains only the proof needed for the coefficient above:

1. constant-width identities and minimum-circumsphere contacts;
2. the lower-circumradius radial certificate;
3. affine contact coordinates and determinant transfer;
4. the high-resolution forced fan;
5. the dominant-edge coordinate-width estimate;
6. six optimized pairwise-disjoint antipodal cone caps;
7. universal branch assembly; and
8. the certificate architecture and trust boundary.

Later private circumradius-local and contact-atlas research is not included. The manuscript contains only a general sentence noting that further refinements may be investigated.

Visual review

The reference PDF was rendered page by page during construction and checked for clipped text, overlapping tables, broken glyphs, and missing references. The release gate still requires final author review of every page after citation, license, version, repository URL, and affiliation metadata are fixed.